Zachary Yedidia

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Research interests: operating systems, security, compilers, computer architecture.

EDUCATION

Stanford University — Stanford, CA
Ph.D. candidate in Computer Science — *Sep. 2021 – June 2027 (expected)*
Advisor: Professor David Mazières

Harvard University — Cambridge, MA
S.M. in Computer Science — *Sep. 2020 – May 2021*

Harvard College — Cambridge, MA
A.B. Magna cum laude with highest honors in Computer Science — *Aug. 2017 – May 2021*

PUBLICATIONS

LFI-XL: Scaling Software-Based Sandboxing to Modern Applications — **IEEE S&P 2027**
Zachary Yedidia, Abhishek Sharma, Anand Balaji, Taehyun Noh, Tal Garfinkel, Hovav Shacham, Shravan Narayan — To appear

Mohabi: Disaggregating and Sandboxing the Firefox JavaScript Engine — **OSDI 2026**
Abhishek Sharma, Anand Balaji, Zachary Yedidia, Anthony Du, Taehyun Noh, Iain Ireland, Jan de Mooij, Matthew Gaudet, Tal Garfinkel, Deian Stefan, Hovav Shacham, Shravan Narayan
Honorable mention for best paper award

Automated Formal Verification of a Software Fault Isolation System (short paper) — **FMCAD 2025**
Matthew Sotoudeh and Zachary Yedidia

Deterministic Client: Enforcing Determinism on Untrusted Machine Code — **OSDI 2025**
Zachary Yedidia, Geoffrey Ramseyer, David Mazières

Segue & ColorGuard: Optimizing SFI Performance and Scalability on Modern Architectures — **ASPLOS 2025**
Shravan Narayan, Tal Garfinkel, Evan Johnson, Zachary Yedidia, Yingchen Wang, Andrew Brown, Anjo Vahldiek-Oberwagner, Michael LeMay, Wenyong Huang, Xin Wang, Mingqiu Sun, Dean Tullsen, Deian Stefan

DeCl: Deterministic and Metered Native Sandboxes — **SIB 2024**
Zachary Yedidia, Geoffrey Ramseyer, David Mazières — Workshop

Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing — **ASPLOS 2024**
Zachary Yedidia

Fast Incremental PEG Parsing — **SLE 2021**
Zachary Yedidia and Stephen Chong
Best paper award

Precision Batching: Bitserial Decomposition for Efficient Neural Network Inference on GPUs — **PACT 2021**
Maximilian Lam, Zachary Yedidia, Colby Banbury, Vijay Janapa Reddi

OPEN SOURCE PROJECTS

Lightweight Fault Isolation (LFI) — Paper, Code
A state-of-the-art software-based sandboxing system supporting AArch64 and x86-64 that enforces memory isolation on machine code within a single address space. Used to sandbox Firefox's SpiderMonkey JavaScript engine (Mohabi, LFI-XL) and to enforce determinism on untrusted machine code for smart contracts (Deterministic Client). It has been adopted by LLVM, where I am the maintainer, and deployed in Android to isolate third-party audio codec libraries.

LLVM Docs, Code
Maintainer of the LFI feature in LLVM, including the AArch64 and x86-64 LFI targets. 29 commits with 8,500+ lines written.

Micro Text Editor Website, Code
Terminal-based text editor I created in 2016 as a modern successor to Nano. It has 25,000+ GitHub stars, 1,000,000+ downloads, and 300+ contributors. Micro is available in Homebrew, Apt, Arch Linux, Snap, and more, and is the default editor on Garuda Linux. Also wrote Mu, a personal rewrite tailored to my preferences that I use daily.

Arcilator · A hardware simulator that I helped develop as part of the CIRCT project while at SiFive. Code
GPeg · Incremental PEG parsing library from my senior thesis with Prof. Stephen Chong. Paper, Thesis, Slides, Code
Literate · A tool for compiling Literate programs written in any programming language. Website, Code
Eget · A tool for automatically installing pre-built binaries distributed in GitHub releases. Code
Go Generic Data Structures · A library of generic data structures for Go. Code
Knit · A flexible build tool that combines Lua with Make's declarative rules language. Code
Multiplix · A small operating system for AArch64 and RISC-V SBCs written in D. Code
Riscinator · A simple 3-stage pipelined RISC-V core and SoC implemented in Chisel. Code
SystemVerilog Guide · Course materials for CS 141 (Spring 2020) at Harvard. PDF

EXPERIENCE

Google · Student Researcher (Deploying LFI in Android) Mountain View, CA · *Summer 2025 – Present*
SiFive · Compiler Intern (Chisel and CIRCT) San Mateo, CA · *Summer 2022*
Zero ASIC · Intern (Silicon Compiler) Cambridge, MA · *Summer 2021*
Harvard University · HCRP Research Fellow (Advisor: Prof. Stratos Idreos) Cambridge, MA · *Summer 2020*
Princeton University · Research Assistant (Advisor: Prof. Naveen Verma) Princeton, NJ · *Summer 2019*
Harvard University · PRISE Research Fellow (Advisor: Prof. Eddie Kohler) Cambridge, MA · *Summer 2018*

HONORS

Honorable mention for best paper award, OSDI 2026 (Mohabi).
NSF Graduate Research Fellowship (2022–2024).
Best paper award, SLE 2021 (Fast Incremental PEG Parsing).
Derek Bok Center teaching award, Harvard CS61 (2020).
Derek Bok Center teaching award, Harvard CS141 (2019).

SERVICE

Organizer of Stanford's weekly TGIF CS Department social event (Autumn 2024 — Present).
External reviewer for ACM TOPLAS (2021).

TEACHING

Operating Systems (CS112/CS212) · CA for Prof. David Mazières Stanford · *Spring 2026*
Distributed Systems (CS244b) · CA for Prof. David Mazières Stanford · *Spring 2024*
Advanced Topics in Operating Systems (CS240) · CA for Profs David Mazières and Dawson Engler Stanford · *Spring 2023*
Using Bits to Control Atoms (CS49n) · CA for Prof. Dawson Engler Stanford · *Autumn 2021*
Systems Programming and Machine Organization (CS61) · TF for Profs Eddie Kohler and Minlan Yu Harvard · *Fall 2020*
Computing Hardware (CS141) · TF for Profs David Brooks and Vijay Reddi Harvard · *Spring 2019, Spring 2020*
Compilers (CS153) · TF for Prof. Stephen Chong Harvard · *Fall 2019*

TALKS

LFI-XL: Scaling Software Fault Isolation Beyond 4GiB · Bay Area Security Day *Oct. '26*
Lightweight Fault Isolation in LLVM, One Year Later · LLVM Developers' Meeting (Quick Talk) *Oct. '26*
Lightweight Fault Isolation in LLVM: Current Support and Future Directions · LLVM Memory Safety Workshop *Oct. '26*
Kage: device driver isolation in the Linux kernel with LFI · Linux Plumbers Conference (Toolchains Track) *Oct. '26*
Securing Android's Audio Codecs with Lightweight Fault Isolation · Google LLVM Summit *Aug. '26*

Mohabi: Disaggregating and Sandboxing the Firefox JavaScript Engine · ORCA Tech Talk (with Sharma, Balaji) *Aug. '26*
Enabling Bare-Metal Smart Contracts With Deterministic Client · Stanford IORH Mini-Conference *July '26*
Quark: Compiler Extensions for Post-Link Instrumentation · Stanford Security Workshop *Apr. '26*
Enabling Bare-Metal Smart Contracts With Deterministic Client · Stanford FDCI Seminar *Mar. '26*
Lightweight Fault Isolation: From Research to Production Sandboxing · Stanford Security Lunch *Mar. '26*
Lightweight Fault Isolation: LLVM Support for Efficient Native Code Sandboxing · LLVM Developers' Meeting *Oct. '25*
Lightweight Fault Isolation: LLVM Support for Efficient Native Code Sandboxing · LLVM Memory Safety WG *Sep. '25*
Securing Native Libraries on Android with Lightweight Fault Isolation (LFI) · Google LLVM Summit *Aug. '25*
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · OSDI *July '25*
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · MIT DCI *May '25*
Securing Native Libraries on Android with LFI · Qualcomm Product Security Summit *May '25*
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · Stanford Security Workshop *Apr. '25*
Lightweight Fault Isolation · Software Compartmentalization Community Tech Talk *Apr. '25*
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · Stanford Security Lunch *Mar. '25*
Sandboxing Native Libraries on Android with Lightweight Fault Isolation (LFI) · Google Android Bootcamp *Feb. '25*
DeCl: Deterministic and Metered Native Sandboxes · SIB Workshop *Sep. '24*
DeCl: Deterministic and Metered Native Sandboxes · Stanford Software Lunch *July '24*
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing · ASPLOS *Apr. '24*
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing · Stanford Security Workshop *Apr. '24*
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing · Stanford Security Lunch *Nov. '23*
Classic Software Fault Isolation and WebAssembly · WebAssembly Research Day *Oct. '23*
Multiplix: using D for kernel development · DConf *Sep. '23*
Fast Incremental PEG Parsing · SLE *Nov. '21*
Fast Incremental PEG Parsing · Stanford Software Lunch *Nov. '21*

POSTERS

Beyond Bounds Checks: Reducing Overheads in Embedded WebAssembly **OSDI 2026**
Nicole Mulvey, Leon Schuermann, Zachary Yedidia, Philip Levis

Lightweight Fault Isolation for RISC-V **RISC-V Summit 2025**
Nicole Mulvey and Zachary Yedidia

Link-time Instrumentation with Quark **OSDI 2025**
Michael Paper and Zachary Yedidia

Stateless SFI: Safer and Faster Machine Code Validation **OSDI 2024**
Zachary Yedidia, Matthew Sotoudeh, David Mazières

Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing **ASPLOS 2024**
Zachary Yedidia

Using Debug Hardware to Build Tiny and Effective Dynamic Kernel Checkers **SOSP 2023**
Zachary Yedidia, Akshay Srivatsan, Manya Bansal, Dawson Engler